



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Smart battery sensor

Advanced AI applications like computer vision and language models rely on cloud data centres, causing latency, cost, security risks, and dependence on connectivity. EnCharge AI's EN100



solves this by enabling AI inference on laptops, workstations, and edge devices, delivering over 200 TOPS of compute power within energy limits. It offers up to 20 times better performance per watt compared to similar solutions, supports up to 128GB memory with a bandwidth of 272GB/s, and handles AI tasks that typically require data centre hardware. Its programmability ensures it adapts to AI models as they evolve, helping developers and device makers build faster, more efficient, and secure applications at the edge.

EnChargeAI

<https://en100.enchargeai.com>

ARM-based lookaside acceleration solution

Mobile network operators face challenges in deploying Open RAN networks due to high energy consumption, limited flexibility, and rising



costs, which make it harder to expand and operate networks. To address these issues, SynaXG and Kalray have introduced an

ARM-based lookaside acceleration solution that combines Kalray's K300 PCIe lookaside acceleration card, powered by the Coolidge v2 MPPA processor, with SynaXG's O-RAN software. This containerised, cloud-deployable solution delivers energy-efficient,

scalable, and real-time acceleration, reducing power use while supporting cost-effective Open RAN deployments. By enabling software-defined Open RAN with the separation of software and hardware, it allows operators to expand networks more easily, deploy 5G applications, and improve network performance without relying on expensive, power-heavy infrastructure.

KALRAY and SynaXG

<https://www.kalrayinc.com> and

<https://www.synaxg.com>

Long-range wireless SoC

Silicon Labs' FG23L addresses the challenge of providing long-range, secure, and energy-efficient IoT connectivity at lower cost, avoiding trade-offs between range, security, and efficiency. It is designed for industrial automation, smart



cities, building automation, agricultural IoT, and electronic shelf labels, where reliable, long-term connectivity is needed. With a link budget of 146dB, +20dBm transmit power, high receiver sensitivity, and low power consumption, the FG23L provides up to twice the range of comparable devices and over 10 years of battery life, reducing maintenance and total cost of ownership. By combining performance, energy efficiency, and cost-effectiveness, manufacturers can scale IoT deployments and design connected products that work in cost-sensitive markets.

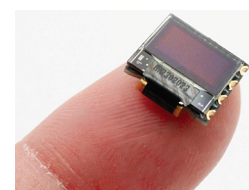
Silicon Labs

<https://news.silabs.com>

OLED display

As electronics get smaller, displaying information becomes a challenge, and

standard solutions are often too large for compact devices like wearables, small instruments, or tiny gadgets. The 8.13mm (0.32-inch) OLED display solves this problem by offering a screen in a small size, 20% smaller than the previous 10.67mm (0.42-inch) OLED



display. It is used in gaming consoles to display scores, level indicators, or character

information, and in wearable devices like smartwatches, fitness trackers, or augmented reality glasses to show notifications, time, or health data. In IoT projects, sensor readings, control status, or real-time information, such as temperature, humidity, or power usage, are displayed. This is also applicable to portable instruments like DIY multimeters, oscilloscopes, or environmental sensors. It can also be used in wearable art or jewellery, where it can display animations, patterns, or messages, enabling developers and makers to build compact, low-power devices that still show information.

Questwise Ventures in the USA

<https://www.questwiseventures.com>

RISC-V CPUs

AI workloads are becoming more complex, and edge devices face limits in processing power, energy use, and flexibility, making it difficult to run AI efficiently. SiFive's 2nd generation



intelligence family addresses these challenges with five RISC-V-based products designed to accelerate AI across a range of applications. The IP